

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 7: Eigensystems

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October 7, 2026

1 Keywords

- Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Condition Number, Rank, Low Rank Approximation, Non-Normal Matrices

2 Main Points

2.1 Eigen-Systems

- We can understand ill-conditioning for linear regression by the eigen-decomposition of $\mathbf{M} = \mathbf{X}^T \mathbf{X}$
- This should be revision
- You know that an *eigenvector*, v , satisfies $\mathbf{M} v = \lambda v$
- For a symmetric matrix there are n real orthogonal eigenvectors
- You can prove they are orthogonal
- **Orthogonal Matrices**
 - Putting the n eigenvectors into a matrix $\mathbf{V}$ with columns v_i we obtain an orthogonal matrix
 - The defining property of an orthogonal matrix is $\mathbf{V}^T \mathbf{V} = \mathbf{V} \mathbf{V}^T = \mathbf{I}$
 - They correspond to rotations (with a possible reflection): they preserve lengths and angles, since $(\mathbf{V}x)^T (\mathbf{V}y) = x^T \mathbf{V}^T \mathbf{V} y = x^T y$
 - $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ follows directly from the eigenvectors being orthonormal. $\mathbf{V} \mathbf{V}^T = \mathbf{I}$ then follows because $\mathbf{V}$ is invertible: if $\mathbf{V}x = 0$ then $x = \mathbf{V}^T \mathbf{V}x = 0$
- **Matrix Decomposition**
 - We can decompose a symmetric matrix, $\mathbf{M}$ as

$$\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$$

- Where $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues of $\mathbf{M}$ (i.e. $\Lambda_{ii} = \lambda_i$)
- $\mathbf{V}$ is the orthogonal matrix made up of the eigenvectors of $\mathbf{M}$
- We can interpret the mapping of a symmetric matrix $\mathbf{M}$ as equivalent to
 1. a rotation defined by $\mathbf{V}^T$
 2. scaling of the i^{th} component by λ_i and
 3. a rotation backwards given by $\mathbf{V}$

- Equivalently if we work in a coordinate system defined by the eigenvectors of $\mathbf{M}$ (this forms an orthonormal basis set) then we just rescale in the directions $\mathbf{v}_i$ by λ_i
 - A symmetric matrix just squashes or expands in different orthogonal directions (this is what the eigensystem captures)

• Inverse Matrices

- The inverse of a symmetric matrix $\mathbf{M}$ is given by

$$\mathbf{M}^{-1} = \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T$$

- Where $\mathbf{\Lambda}^{-1}$ is a diagonal matrix with elements $\Lambda_{ii}^{-1} = 1/\lambda_i$
- This is only defined if all the eigenvalues of $\mathbf{M}$ are non-zero ($\mathbf{M}$ is said to be full rank)
- If λ_i is very small then $1/\lambda_i$ is large and in taking the inverse $\mathbf{M}^{-1}\mathbf{x}$ any component of $\mathbf{x}$ in the direction $\mathbf{v}_i$ will get magnified by $1/\lambda_i$
- For linear regression we invert $\mathbf{M} = \mathbf{X}^T \mathbf{X}$
 - in directions where the training examples don't vary much the associated eigenvalue will be small and the inverse inherently unstable

2.2 Condition Number

- Suppose we solve $\mathbf{M}\mathbf{x} = \mathbf{b}$ for a symmetric, invertible $\mathbf{M}$, but $\mathbf{b}$ carries an error $\delta\mathbf{b}$. The solution then carries an error $\delta\mathbf{x} = \mathbf{M}^{-1}\delta\mathbf{b}$
- In the eigenvector basis $\mathbf{M}^{-1}$ just multiplies the component along $\mathbf{v}_i$ by $1/\lambda_i$, so

$$\|\delta\mathbf{x}\| \leq \frac{\|\delta\mathbf{b}\|}{|\lambda_{\min}|} \quad \text{and} \quad \|\mathbf{b}\| = \|\mathbf{M}\mathbf{x}\| \leq |\lambda_{\max}| \|\mathbf{x}\|$$

Rearranging the second as $1/\|\mathbf{x}\| \leq |\lambda_{\max}|/\|\mathbf{b}\|$ and multiplying it by the first gives

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}, \quad \kappa = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

- κ is the *condition number*. A relative error in the data can be amplified by up to a factor κ in the solution
- For a symmetric matrix $|\lambda_{\max}|$ is the spectral norm $\|\mathbf{M}\|_2$ from the vector spaces lecture, and $|\lambda_{\min}|^{-1} = \|\mathbf{M}^{-1}\|_2$, so $\kappa = \|\mathbf{M}\|_2 \|\mathbf{M}^{-1}\|_2$
- Example: in the mappings lecture $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1.001 \end{pmatrix}$ has eigenvalues ≈ 2.0005 and 0.0005 , so $\kappa \approx 4000$. Changing y_2 by 0.001 is a relative change of 3.5×10^{-4} in $\mathbf{y}$, but changed $\mathbf{w}$ from $(1, 1)$ to $(0, 2)$, a relative change of 1: an amplification of about 2800, within the bound

2.3 Rank and Null Space

- Writing out $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$ column by column gives the useful form

$$\mathbf{M} = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

a sum of n rank-one pieces, one per eigenvector

- The *rank* of a symmetric matrix is its number of non-zero eigenvalues. (For a general square matrix the rank is the number of linearly independent columns; the two agree for symmetric matrices but not in general)

- The eigenvectors with zero eigenvalue span the *null space*: any $x = \sum_k a_k v_k$ built from them has $\mathbf{M}x = 0$
- A matrix is *rank deficient* if it has a zero eigenvalue. Equivalently its columns are linearly dependent, or $\det(\mathbf{M}) = \prod_i \lambda_i = 0$. It then has no inverse: if x solves $\mathbf{M}x = b$ then so does x plus anything in the null space

2.4 Pseudo-Inverse of a Symmetric Matrix

- If $\mathbf{M}$ has k non-zero eigenvalues $\lambda_1, \dots, \lambda_k$ we define

$$\mathbf{M}^+ = \mathbf{V} \mathbf{\Lambda}^+ \mathbf{V}^T = \sum_{i=1}^k \frac{1}{\lambda_i} \mathbf{v}_i \mathbf{v}_i^T, \quad \mathbf{\Lambda}^+ = \text{diag}\left(\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0\right)$$

i.e. invert the non-zero eigenvalues and leave the zero ones at zero

- If $\mathbf{M}x = b$ has solutions, $x = \mathbf{M}^+b$ is the one with no component in the null space. Since every other solution adds a null-space vector, which is orthogonal to it, this is the solution with the smallest $\|x\|$
- If b has a component in the null space there is no exact solution, and $\mathbf{M}^+b$ minimises $\|\mathbf{M}x - b\|$ instead
- This is the symmetric case of the Moore–Penrose pseudo-inverse described in the SVD notes

2.5 Low Rank Approximation

- Keeping only the k largest (in magnitude) eigenvalues gives the rank- k approximation $\mathbf{M}_k = \sum_{i=1}^k \lambda_i \mathbf{v}_i \mathbf{v}_i^T$
 - It needs only k eigenvalues and k eigenvectors, $k(n+1)$ numbers rather than n^2
 - The error, $\mathbf{M} - \mathbf{M}_k$, is made of the discarded terms, so it is small when the discarded eigenvalues are small
- For an ill-conditioned matrix we can set the tiny eigenvalues to zero and use the pseudo-inverse. The directions that would have amplified noise by $1/\lambda_i$ are simply ignored
 - This trades a little bias (we lose whatever true signal lay in those directions) for a large reduction in variance
 - The experiment below shows it in action. Ridge regression, in the SVD lecture, is a smoother version of the same idea: it damps small eigenvalues rather than discarding them

2.6 Non-Normal Matrices

- Symmetric matrices belong to a larger class, the *normal* matrices, which satisfy $\mathbf{M} \mathbf{M}^\dagger = \mathbf{M}^\dagger \mathbf{M}$, where $\mathbf{M}^\dagger = (\mathbf{M}^T)^*$ is the conjugate transpose (for real matrices just $\mathbf{M}^T$)
 - Normal matrices are exactly the matrices with a full set of orthonormal (possibly complex) eigenvectors
 - Hermitian matrices ($\mathbf{M} = \mathbf{M}^\dagger$) are normal; they matter in quantum theory, much less in machine learning
 - A real matrix can be normal without being symmetric: the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is normal, with eigenvalues $\pm i$

- Most non-symmetric real matrices are not normal, and their eigenvectors behave less nicely
- **Complex eigenvalues.** The eigenvalues are the roots of the *characteristic equation* $\det(\mathbf{M} - \lambda \mathbf{I}) = 0$. $(\mathbf{M} - \lambda \mathbf{I}) \mathbf{v} = \mathbf{0}$ can only have a non-zero solution if $\mathbf{M} - \lambda \mathbf{I}$ is singular. This is a polynomial of order n in λ , so there are n roots (counting repeats), and in general they are complex
 - If $\mathbf{M}$ is real and λ is complex then $\mathbf{v}$ is complex too. Taking the complex conjugate of $\mathbf{M} \mathbf{v} = \lambda \mathbf{v}$ gives $\mathbf{M} \mathbf{v}^* = \lambda^* \mathbf{v}^*$, so complex eigenvalues and eigenvectors come in conjugate pairs
- **Defective matrices.** A non-symmetric matrix need not have n linearly independent eigenvectors; if it does not it is called *defective*
 - For example $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the repeated eigenvalue 1 but only one eigenvector, $(1, 0)^T$
 - A randomly generated matrix is almost never defective, but defective matrices do turn up in structured problems
- **Left and right eigenvectors.** As well as the usual (right) eigenvectors $\mathbf{M} \mathbf{v}_i = \lambda_i \mathbf{v}_i$ there are left eigenvectors $\mathbf{u}_i^T \mathbf{M} = \lambda_i \mathbf{u}_i^T$, i.e. eigenvectors of $\mathbf{M}^T$. As $\det(\mathbf{M}^T - \lambda \mathbf{I}) = \det(\mathbf{M} - \lambda \mathbf{I})$ they have the same eigenvalues
 - For a symmetric matrix the left eigenvectors are just the transposes of the right ones
 - In general they differ, and the right eigenvectors are not orthogonal to one another. Instead the two sets are *biorthogonal*: computing $\mathbf{u}_i^T \mathbf{M} \mathbf{v}_j$ in two ways (exactly as in the proof for symmetric matrices) gives $(\lambda_i - \lambda_j) \mathbf{u}_i^T \mathbf{v}_j = 0$, so $\mathbf{u}_i^T \mathbf{v}_j = 0$ whenever $\lambda_i \neq \lambda_j$
- **Eigen-decomposition.** If $\mathbf{M}$ is not defective, put the right eigenvectors in the columns of $\mathbf{V}$. Then $\mathbf{M} \mathbf{V} = \mathbf{V} \mathbf{\Lambda}$ and, as the columns are independent, $\mathbf{V}$ is invertible, so

$$\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$$

- $\mathbf{V}$ is no longer orthogonal, so $\mathbf{V}^{-1} \neq \mathbf{V}^T$
- Multiplying $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$ on the left by $\mathbf{V}^{-1}$ gives $\mathbf{V}^{-1} \mathbf{M} = \mathbf{\Lambda} \mathbf{V}^{-1}$: the rows of $\mathbf{V}^{-1}$ are the left eigenvectors, scaled so that $\mathbf{u}_i^T \mathbf{v}_i = 1$
- We still get $\mathbf{M}^k = \mathbf{V} \mathbf{\Lambda}^k \mathbf{V}^{-1}$, but the geometric picture (rotate, stretch along orthogonal axes, rotate back) is lost. This is one reason we lean on the singular value decomposition, which works for every matrix

3 Exercises

1. Let $\mathbf{M}$ be a symmetric matrix with eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ and distinct eigenvalues $\lambda_1 \neq \lambda_2$. By computing $\mathbf{v}_2^T \mathbf{M} \mathbf{v}_1$ in two different ways, show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal
2. Show that the eigenvalues of $\mathbf{X}^T \mathbf{X}$ are never negative. (Hint: consider $\|\mathbf{X} \mathbf{v}\|^2$)
3. Using $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$
 - (a) Show that $\mathbf{M}^{-1} = \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T$ by multiplying the two together
 - (b) Show that $\mathbf{M}^k = \mathbf{V} \mathbf{\Lambda}^k \mathbf{V}^T$
4. Find the eigenvalues and eigenvectors of $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and check that $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$
5. Non-normal matrices

- (a) For $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ find the eigenvalues, the right eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ and the left eigenvectors $\mathbf{u}_1, \mathbf{u}_2$. Check that $\mathbf{v}_1^\top \mathbf{v}_2 \neq 0$ but $\mathbf{u}_1^\top \mathbf{v}_2 = \mathbf{u}_2^\top \mathbf{v}_1 = 0$
- (b) Verify that $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$ and that the rows of $\mathbf{V}^{-1}$ are the left eigenvectors
- (c) Show that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one independent eigenvector
- (d) Show that $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is normal but not symmetric, and that $(1, -i)^\top$ is an eigenvector with eigenvalue i . What is the other eigenvector?
6. Show that $\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top = \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$, and use the sum form to check that $\mathbf{M} \mathbf{v}_j = \lambda_j \mathbf{v}_j$
7. For $\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$
- (a) Find the eigenvalues, eigenvectors, rank and null space
- (b) Find $\mathbf{M}^+$
- (c) Solve $\mathbf{M} \mathbf{x} = (2, 2)^\top$. Write down all the solutions and show $\mathbf{M}^+(2, 2)^\top$ is the shortest
- (d) What does $\mathbf{M}^+$ give for $\mathbf{b} = (1, 3)^\top$, which has no exact solution?
8. Work through the derivation of the condition-number bound $\|\delta \mathbf{x}\| / \|\mathbf{x}\| \leq \kappa \|\delta \mathbf{b}\| / \|\mathbf{b}\|$. Find a choice of $\mathbf{b}$ and $\delta \mathbf{b}$ for which it is an equality

4 Experiments

4.1 Eigensystems

- In either Matlab/Octave or python generate random matrices and check the matrix identities

```

X = randn(5,4) % generate a mock design matrix with 5 inputs of length 4
M = X'*X      % compute a symmetric matrix
[V,L] = eig(M) % compute eigenvectors and eigenvalues
V*L*V'        % should be identical to M
V*V'          % should be the identity matrix (up to rounding precision)
V'*V          % should be the identity matrix (up to rounding precision)
x = randn(4,1) % generate a random column matrix of length 4
y = randn(4,1) % generate another random column matrix of length 4
xp = V*x       % apply V to x
yp = V*y       % apply V to y
norm(x)        % compute Euclidean norm of x
norm(xp)       % should be the same as the Euclidean norm of x
x'*y           % compute inner product of x and y
xp'*yp         % inner product of xp and yp (should be the same as above)

Z = rand(4,5)  % a design matrix with more unknowns than examples
W = Z'*Z      % a covariance-like matrix (but without subtracting the mean)
eig(W)        % compute eigenvalues (one should be 0 up to machine precision)

```

4.2 Non-Normal Matrices

- In python, look at the eigensystem of a random non-symmetric matrix

```

import numpy as np

rng = np.random.default_rng(2)
M = rng.normal(size=(4, 4))          # not symmetric
lam, V = np.linalg.eig(M)            # columns of V are right eigenvectors
print(lam)                           # complex eigenvalues come in conjugate pairs
print(np.allclose(M, V @ np.diag(lam) @ np.linalg.inv(V))) # M = V L V^-1
U = np.linalg.inv(V)                 # rows should be left eigenvectors
print(np.round(U @ V, 10))           # u_i^T v_j: the identity (biorthogonal)
print(np.round(V.conj().T @ V, 3))   # but the v_i are not orthogonal

```

4.3 Low Rank Pseudo-Inverse

- Regression with two almost identical features: compare inverting every eigenvalue of $\mathbf{X}^T \mathbf{X}$ with dropping the tiny one

```

import numpy as np

rng = np.random.default_rng(0)
m = 50
X = rng.normal(size=(m, 3))
X[:, 2] = X[:, 1] + 1e-3 * rng.normal(size=m) # feature 3 is almost feature 2
w_true = np.array([1.0, 2.0, 0.0])
C = X.T @ X                                     # symmetric 3x3
lam, V = np.linalg.eigh(C)                     # eigenvalues in ascending order
print(lam, lam[-1] / lam[0])                   # tiny eigenvalue; condition no.
for trial in range(3):                         # same X, fresh noise each time
    y = X @ w_true + 0.1 * rng.normal(size=m)
    w_full = V @ np.diag(1 / lam) @ V.T @ X.T @ y # invert all eigenvalues
    lam_plus = np.where(lam > 1e-2 * lam[-1], 1 / lam, 0) # drop the tiny one
    w_trunc = V @ np.diag(lam_plus) @ V.T @ X.T @ y
    print(np.round(w_full, 2), np.round(w_trunc, 2))

```

- The condition number is about 3×10^6 . With every eigenvalue inverted the last two weights jump around from one noise sample to the next (e.g. (6.27, -4.26) then (-1.38, 3.36)), though their sum stays near 2. Dropping the tiny eigenvalue gives about (1, 1) every time
- Neither recovers the true (2, 0): the data cannot tell the two features apart. The truncated answer splits the weight equally, the minimum-norm choice, and predicts just as well

5 Answers

1. On one hand $\mathbf{v}_2^T (\mathbf{M} \mathbf{v}_1) = \lambda_1 \mathbf{v}_2^T \mathbf{v}_1$. On the other, as $\mathbf{M}^T = \mathbf{M}$, $\mathbf{v}_2^T \mathbf{M} \mathbf{v}_1 = (\mathbf{M} \mathbf{v}_2)^T \mathbf{v}_1 = \lambda_2 \mathbf{v}_2^T \mathbf{v}_1$. Subtracting, $(\lambda_1 - \lambda_2) \mathbf{v}_2^T \mathbf{v}_1 = 0$ and since $\lambda_1 \neq \lambda_2$ we must have $\mathbf{v}_2^T \mathbf{v}_1 = 0$
2. If $\mathbf{X}^T \mathbf{X} \mathbf{v} = \lambda \mathbf{v}$ then $\lambda \|\mathbf{v}\|^2 = \mathbf{v}^T \mathbf{X}^T \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 \geq 0$, so $\lambda \geq 0$. It is zero exactly when $\mathbf{X} \mathbf{v} = \mathbf{0}$: a direction the data say nothing about
3. Matrix powers
 - (a) $\mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \mathbf{V} \mathbf{\Lambda}^{-1} \mathbf{V}^T = \mathbf{V} \mathbf{\Lambda} \mathbf{\Lambda}^{-1} \mathbf{V}^T = \mathbf{V} \mathbf{V}^T = \mathbf{I}$, using $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ twice
 - (b) In $\mathbf{M}^k = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \dots \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$ every inner $\mathbf{V}^T \mathbf{V}$ is $\mathbf{I}$, leaving $\mathbf{V} \mathbf{\Lambda}^k \mathbf{V}^T$

4. $\det(\mathbf{M} - \lambda \mathbf{I}) = (2 - \lambda)^2 - 1 = 0$ gives $\lambda = 3$ with $\mathbf{v} = (1, 1)^\top / \sqrt{2}$ and $\lambda = 1$ with $\mathbf{v} = (1, -1)^\top / \sqrt{2}$. Then

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = \mathbf{M}$$

5. Non-normal matrices

- (a) $\mathbf{M}$ is upper triangular, so $\det(\mathbf{M} - \lambda \mathbf{I}) = (1 - \lambda)(3 - \lambda)$ and the eigenvalues are 1 and 3. Right eigenvectors: $\mathbf{v}_1 = (1, 0)^\top$ and $\mathbf{v}_2 = (1, 1)^\top$. Left eigenvectors (eigenvectors of $\mathbf{M}^\top$): $\mathbf{u}_1 = (1, -1)^\top$ and $\mathbf{u}_2 = (0, 1)^\top$. Then $\mathbf{v}_1^\top \mathbf{v}_2 = 1 \neq 0$, while $\mathbf{u}_1^\top \mathbf{v}_2 = 1 - 1 = 0$ and $\mathbf{u}_2^\top \mathbf{v}_1 = 0$. (With these choices $\mathbf{u}_1^\top \mathbf{v}_1 = \mathbf{u}_2^\top \mathbf{v}_2 = 1$ already.)

- (b) $\mathbf{V} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathbf{V}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, whose rows are $\mathbf{u}_1^\top$ and $\mathbf{u}_2^\top$, and

$$\mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \mathbf{M}$$

- (c) The only eigenvalue is 1 (twice). An eigenvector must satisfy $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0}$, i.e. $v_2 = 0$, so every eigenvector is a multiple of $(1, 0)^\top$

- (d) $\mathbf{R} \mathbf{R}^\top = \mathbf{R}^\top \mathbf{R} = \mathbf{I}$ (it is a rotation by 90°), but $\mathbf{R}^\top \neq \mathbf{R}$. $\mathbf{R} (1, -i)^\top = (i, 1)^\top = i (1, -i)^\top$. The other eigenvector is the complex conjugate, $(1, i)^\top$, with eigenvalue $-i$. As $\mathbf{R}$ is normal these two are orthogonal, using the complex inner product (conjugate the first vector): $(1, i)^\dagger (1, -i)^\top = 1 \cdot 1 + (-i)(-i) = 1 - 1 = 0$

6. $\mathbf{V} \mathbf{\Lambda}$ has columns $\lambda_i \mathbf{v}_i$, and a matrix with columns $\mathbf{a}_i$ times one with rows $\mathbf{v}_i^\top$ is $\sum_i \mathbf{a}_i \mathbf{v}_i^\top$. So $\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top = \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$. Then $\mathbf{M} \mathbf{v}_j = \sum_i \lambda_i \mathbf{v}_i (\mathbf{v}_i^\top \mathbf{v}_j) = \lambda_j \mathbf{v}_j$, since $\mathbf{v}_i^\top \mathbf{v}_j = 0$ unless $i = j$

7. Rank-deficient example

- (a) $\lambda = 2$ with $\mathbf{v}_1 = (1, 1)^\top / \sqrt{2}$, and $\lambda = 0$ with $\mathbf{v}_2 = (1, -1)^\top / \sqrt{2}$. The rank is 1 and the null space is all multiples of $(1, -1)^\top$

(b) $\mathbf{M}^+ = \frac{1}{2} \mathbf{v}_1 \mathbf{v}_1^\top = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

- (c) The solutions are $x_1 + x_2 = 2$, i.e. $(1, 1)^\top + t(1, -1)^\top$ for any t , with $\|\mathbf{x}\|^2 = 2 + 2t^2$. This is smallest at $t = 0$, and $\mathbf{M}^+(2, 2)^\top = \frac{1}{4}(4, 4)^\top = (1, 1)^\top$

- (d) $\mathbf{M}^+(1, 3)^\top = \frac{1}{4}(4, 4)^\top = (1, 1)^\top$, giving $\mathbf{M} \mathbf{x} = (2, 2)^\top$. That is the closest point to $(1, 3)^\top$ that $\mathbf{M}$ can reach: the projection of $(1, 3)^\top$ onto $(1, 1)^\top$

8. Equality needs both inequalities to be tight: $\mathbf{x}$ along the eigenvector of $\lambda_{\max}$, so that $\|\mathbf{M} \mathbf{x}\| = |\lambda_{\max}| \|\mathbf{x}\|$, and $\delta \mathbf{b}$ along the eigenvector of $\lambda_{\min}$. So take $\mathbf{b} = \mathbf{v}_{\max}$ and $\delta \mathbf{b} = \epsilon \mathbf{v}_{\min}$: the signal lies in the direction $\mathbf{M}$ stretches most and the noise in the one it stretches least